NetGuardian Security Report

Reporting Period: All Time / Full Operational Session · Generated: 2026-11-11 14:45:48 UTC · Mode: SIMULATION

SIMULATED DATA NOTICE: All network telemetry, topology nodes, traffic flows, and security alerts in this report are generated via the NetGuardian Simulation Engine for defensive cybersecurity research and training.

1. Network Overview & Health Score

Network Health: 100.0% (HEALTHY)	**Devices Monitored:** 12	**Devices Online:** 12
Active Threats: 0	**AI Anomalies Flagged:** 0	**Isolated Devices:** 0

Formula Component	Impact	Description
Baseline Health	100.0 pts	Clean network baseline standard.
Active Threat Penalty	-0.0 pts	Penalty applied for uncontained alerts and open threats.
AI Anomaly Penalty	-0.0 pts	Penalty applied for active Isolation Forest behavioral outliers.
Unregistered Device Penalty	-0.0 pts	Penalty applied for rogue / unregistered network devices.
Containment Credit	+0.0 pts	Credit awarded for isolated endpoints mitigating lateral risk.
Effective Network Health	**100.0% (HEALTHY)**	**Health = max(0, 100 - Active Threats - Suspicious Devices - High-Risk Nodes - Anomalies - Offline Devices + Credit)**

2. Monitored Devices Inventory

Name	IP Address	Type	Criticality	Risk	Status
Internet	203.0.113.1	Internet	MEDIUM	0.0	ONLINE
Router	192.168.1.1	Router	HIGH	0.0	ONLINE
Switch	192.168.1.2	Switch	HIGH	0.0	ONLINE
PC-01	192.168.1.10	Workstation	MEDIUM	0.0	ONLINE
PC-02	192.168.1.11	Workstation	MEDIUM	0.0	ONLINE
PC-03	192.168.1.12	Workstation	MEDIUM	0.0	ONLINE
PC-04	192.168.1.13	Laptop	MEDIUM	0.0	ONLINE
Phone-01	192.168.1.14	Mobile	LOW	0.0	ONLINE
Printer	192.168.1.30	Printer	LOW	0.0	ONLINE
Server	192.168.1.20	Server	HIGH	0.0	ONLINE
Database	192.168.1.21	Database	CRITICAL	0.0	ONLINE
IoT-Camera	192.168.1.40	Camera	LOW	0.0	ONLINE

3. Network Traffic Summary

Total Packets Monitored: **0** | Total Bytes: **0** | Anomalous Flows: **0**

4. Security Events & Threat Detections

No events in reporting period.

5. AI-Powered Anomaly Detections

No AI anomalies detected in this reporting period.

6. High-Risk Devices & Critical Assets

No high-risk endpoints identified.

7. Simulated Attack Paths (HYPOTHETICAL)

No hypothetical attack paths simulated in this reporting period.

8. Actionable Recommendations

1. Maintain zero-trust access control lists (ACLs) between general workstation subnets and sensitive database servers.
2. Verify all newly observed MAC/IP addresses against authorized organizational IT inventory prior to registration.
3. Perform scheduled What-If lateral movement simulations to validate proactive containment policies.

9. Conclusion

The NetGuardian defensive security architecture successfully maintains continuous behavioral visibility, combining deterministic rule detection with Isolation Forest Machine Learning anomaly scoring. The graph-based Digital Twin model accurately maps lateral traversal blast radiuses, enabling non-destructive threat isolation and measurable risk reduction.

10. Limitations & Methodological Notice

LIMITATIONS & DISCLAIMER: Machine Learning anomaly detections and Isolation Forest outlier scores reflect statistical deviation from established baseline telemetry; anomalies are not proof of attacks. Anomaly flags require corroboration through rule signatures, log correlation, and forensic verification by a qualified security analyst.